

Grinding Safety Training

Voice Over Script Document

Course Title

Grinding Safety Training

Organization

Hitachi Energy

Target Audience

Employees, operators, technicians, and maintenance personnel involved in grinding activities.

Course Duration

Approximately 20–25 Minutes

Course Mode

E-Learning

Course Introduction

Welcome Note

A grinding wheel can rotate at extremely high speeds.
If it breaks, fragments can fly with great force.

Are you prepared to work safely?

Hi! I'm Spark, your guide to Grinding Safety.

Welcome to the Grinding Safety training in Hitachi Energy. I'll be with you throughout this training to help you stay safe.

In this course, we'll explore the hazards involved in grinding and the safe practices you must follow.

Let's begin!

Course Outline

1. Introduction to Grinding Safety
 2. Grinding Hazards and Risks
 3. Safe Use of Grinding Machines
 4. Grinding Wheel Inspection and Handling
 5. Personal and Workplace Safety
 6. Summary
 7. Assessment
-

Module 1: Introduction to Grinding Safety

1.1 Learning Objectives

Voice Over Script

By the end of this course, you will be able to:

- Identify common grinding hazards and risks
- Use grinding machines safely
- Inspect and handle grinding wheels correctly
- Follow important safety practices

1.2 What is Grinding?

Voice Over Script

Grinding is a process used to remove material from a surface using a rotating abrasive wheel. It helps shape, smooth, and finish components with precision.

You'll find grinding operations in many places such as maintenance workshops, fabrication areas, and production floors.

But here's the important part.

Even though grinding is a common activity, it comes with serious risks.

During operation, the machine produces sparks, noise, heat, and flying particles—all of which can become hazards if not properly controlled.

That's why grinding safety is not just about following instructions—it's about understanding the risks behind every action you take.

The grinding wheel itself is both powerful and fragile. While it can cut through hard materials, it can also break unexpectedly if misused.

Spark Tip

“Even a small crack in a wheel can become a big risk at high speed.”

1.3 Consequences of Unsafe Practices

Voice Over Script

Unsafe grinding practices can result in:

- Injuries from flying particles or broken wheel fragments

- Eye and face injuries
- Hearing damage due to high noise levels
- Damage to equipment and workplace incidents

In severe cases, accidents can even be fatal. This is why it is important to always follow safety guidelines.

1.4 Types of Grinding Machines

Voice Over Script

Before we move ahead, let's quickly look at the common types of grinding machines used in the workplace.

Each is designed for specific tasks—but all carry similar risks if not used safely.

1. Portable Hand Grinder (Angle Grinder)

A handheld tool used for cutting, grinding, and polishing.

It offers flexibility but requires strong control and careful handling.

2. Bench Grinder

A fixed machine used for sharpening and shaping metal.

Maintain proper distance and positioning.

Important: Never use gloves—they can get caught in the wheel.

3. Chop Saw (Cut-off Saw)

Used for cutting hard materials with a rotating wheel.

Ensure the workpiece is firmly clamped to prevent movement or kickback.

4. CNC Grinding Machine

Automated and highly precise, used in production environments.

Risks still exist during setup, maintenance, and tool changes.

5. Other Specialized Grinders

Includes surface, cylindrical, and pedestal grinders.

Each has specific uses, but safety principles remain the same.

Closing Statement

No matter the machine, the hazards are real.

Understanding your equipment is the first step to working safely.

Module 2: Grinding Hazards and Risks

2.1 Understanding Grinding Wheel Risks

Voice Over Script

To stay safe, we need to understand what is happening inside the machine.

A grinding wheel isn't a solid piece of metal; it is a fragile disc held together by bonding agents.

When it spins at high speeds, a force called centrifugal force pulls the wheel outward from the center.

Think of the wheel like a coiled spring. It is holding a massive amount of energy.

If the wheel is weakened by a tiny crack, or if you apply bumping or side pressure that it wasn't built for, that energy is released instantly.

Because the wheel is spinning so fast, it doesn't just crack—it explodes.

The broken pieces fly out like projectiles, traveling fast enough to cause fatal injuries to anyone in the area.

The biggest challenge is that these internal cracks are often invisible to the naked eye.

That is why we never rely on a quick look. We rely on the strict safety checks you are about to learn.

Spark Reminder

“A wheel that looks fine can still be a major risk. Safety isn't just what you see; it's what you know is happening inside.”

Module 3: Safe Use of Grinding Machines

3.1 Selecting the Correct Grinding Wheel

Voice Over Script

Before mounting a wheel, perform this quick Three-Point Check to ensure it's a safe match for your machine.

1. Check the Speed Rating

Every wheel has a limit.

You must ensure the wheel's RPM rating is at or above the grinder's speed.

If a grinder spins at 4,000 RPM, a 3,000 RPM wheel will shatter instantly.

2. Check the Wheel Condition

A correct wheel is a healthy wheel.

Because wheels are fragile, you must reject any wheel that has been dropped, rolled on the floor, or exposed to moisture.

These can cause internal cracks that are invisible to the eye.

3. Check the Application

Always match the wheel to the task.

Never apply side pressure to a thin, straight wheel—they are designed for edge grinding only.

Pushing against the side can cause the wheel to snap.

Spark Tip

“If the RPM doesn't match or the wheel was mistreated, don't risk it. Grab a new one.”

3.2 Safe Operating Practices

Voice Over Script

Before starting, ensure that the grinding wheel guard is properly installed.

Never operate a grinder without a guard.

Check proper positioning and stability.

Maintain proper clearances:

- The gap between the tool rest and the wheel should not exceed 1/8 inch
- The distance between the wheel and the top guard should not exceed 1/4 inch

While grinding:

- Do not apply excessive pressure on the wheel
- Avoid any impact or sudden force on the wheel
- Do not grind on the side of a straight wheel unless it is designed for that purpose
- Stand slightly to the side during startup

Spark Tip

“Take a few seconds to check your setup—it can prevent major accidents.”

These practices help prevent damage to the wheel and reduce the risk of accidents.

Module 4: Grinding Wheel Inspection and Handling

4.1 Importance of Wheel Inspection

Voice Over Script

Always inspect the grinding wheel before use.

Even if there are no visible defects, the wheel may still be damaged internally.

Using a damaged wheel can result in sudden breakage during operation.

Check the wheel carefully for cracks, chips, or any visible damage.

In addition, perform a ring test before mounting the wheel.

To perform the ring test, gently tap the wheel.

- A clear ringing sound indicates that the wheel is in good condition
- A dull sound indicates possible cracks or internal damage

Never use a wheel that produces a dull sound.

4.2 Handling and Storage

Voice Over Script

Grinding wheels can be easily damaged during handling.

Follow these precautions:

- Do not drop or hit the wheel
- Avoid rolling the wheel on the floor
- Protect the wheel from moisture, water, and direct heat
- Store wheels in a dry and safe location

Spark Reminder

“A simple ring test can help you avoid a serious accident.”

Always mount the wheel properly and perform a test run in a safe or enclosed area before actual use.

Module 5: Personal and Workplace Safety

5.1 Personal Protective Equipment (PPE)

Voice Over Script

Always wear appropriate PPE while performing grinding operations.

This includes:

- Safety glasses or a face shield to protect from flying particles
- Hearing protection to reduce noise exposure
- Respiratory protection if dust is present
- Safety shoes and protective clothing

Important: Gloves should not be used while working on bench grinders, as they can get caught in the rotating wheel.

5.2 Pre-Operation and Electrical Safety Checks

Voice Over Script

Before starting:

- Ensure the grinder is properly earthed
- Check all electrical connections
- Verify ELCB (Earth Leakage Circuit Breaker) is working
- Ensure guards and safety devices are in place
- Make sure there are no flammable materials nearby, as sparks can easily start a fire

Also, avoid distractions.

Even something as simple as a mobile phone notification can take your attention away at the wrong moment.

And always be aware of emergency arrangements—know where the exits, fire extinguishers, and first aid kits are located.

Spark Reminder

“Never skip safety checks—they are your first line of defense.”

5.3 Safe Work Practices and Key Safety Reminders

Voice Over Script

Follow these important safety practices at all times:

- Never use a damaged or cracked grinding wheel
- Do not operate the machine without guards
- Avoid applying excessive force
- Stay focused and avoid distractions during operation
- Ensure proper positioning and maintain a safe distance

Always be aware of your surroundings and follow workplace safety procedures.

Module 6: Summary

6.1 Key Takeaways

Voice Over Script

Let's review the key points from this course:

- Grinding involves high-speed rotating wheels that can be dangerous
- Wheel inspection and proper handling are critical
- Safe operating practices reduce the risk of accidents

- PPE and pre-operation checks are essential for personal safety

By following these guidelines, you can create a safer working environment for yourself and your team.

6.2 Closing Slide

Voice Over Script

Thank you for completing this training.

Remember, safety is not just a rule—it is a responsibility.

Always follow safe grinding practices.

Assessment Questions

Q1. What is the main purpose of grinding?

Type

Multiple Choice

Options

- A. To join materials
- B. To remove material and smooth surfaces
- C. To heat materials
- D. To paint surfaces

Correct Answer

B. To remove material and smooth surfaces

Feedback

Correct:

“Correct! Grinding is used to remove material and achieve a smooth finish.”

Incorrect:

“Incorrect. Grinding is mainly used for material removal and finishing.”

Q2. Why is grinding considered dangerous?

Type

Multiple Choice

Options

- A. It produces noise
- B. It uses electricity
- C. Grinding wheels rotate at very high speed
- D. It requires training

Correct Answer

C. Grinding wheels rotate at very high speed

Feedback

Correct:

“Right! High-speed rotation increases the risk of serious accidents.”

Incorrect:

“Not quite. The main risk comes from high-speed rotating wheels.”

Q3. You notice a small crack on a grinding wheel. What should you do?

Type

Scenario-Based MCQ

Options

- A. Use it carefully
- B. Ignore it
- C. Replace the wheel immediately
- D. Apply less pressure

Correct Answer

C. Replace the wheel immediately

Feedback

Correct:

“Correct! Never use a damaged wheel.”

Incorrect:

“Unsafe choice. A cracked wheel must never be used.”

Q4. What does a dull sound in a ring test indicate?

Type

Multiple Choice

Options

- A. Wheel is new
- B. Wheel is safe

- C. Wheel is damaged
- D. Wheel is clean

Correct Answer

- C. Wheel is damaged
-

Q5. What is the correct tool rest gap?

Type

Multiple Choice

Options

- A. 1/2 inch
- B. 1/4 inch
- C. 1/8 inch
- D. No gap

Correct Answer

- C. 1/8 inch